Optima Vision: Clothing-Item Recognition Using Classical Machine Learning

Members :

| Ansh Monga | 590025575 |

| Raunak Singh Negi | 590022626 |

| Prince Sai | 590022756 |

| Anirudh Awasthi | 590027460 |

1. Problem and Objective

Image classification is frequently associated with deep neural networks, but classical machine-learning methods can remain effective when images are represented appropriately. This project investigates how far non-deep learning methods can classify Fashion-MNIST clothing images and examines the computational and classification trade-offs between different model families.

Fashion-MNIST contains 70,000 grayscale clothing images belonging to ten balanced classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag and Ankle boot. Each image has a resolution of 28x28 pixels. The official dataset provides 60,000 training images and 10,000 test images, and this split was preserved throughout the project.

The project compares K-Nearest Neighbours (KNN), an RBF Support Vector Machine (SVM), Random Forest and a small Multilayer Perceptron (MLP). Logistic Regression was additionally used as an initial linear baseline. Principal Component Analysis (PCA) forms the dimensionality-reduction stage shared by the classifiers.

The investigation focuses not only on predictive accuracy but also on macro-F1, training time, prediction time and class-specific errors. This is important because two models with similar accuracy can have substantially different computational costs or weaknesses on individual clothing categories.

2. Data Loading and Exploratory Analysis

The original Fashion-MNIST IDX gzip files were used rather than a library-provided dataset loader. A custom IDX parser was implemented to read the binary headers and pixel/label data. The resulting arrays were verified to have shapes (60000,28,28) and (10000,28,28) for the training and test images respectively. Each of the ten training classes contained 6,000 examples.

Exploratory analysis included labelled sample images, mean images for each class and the overall pixel-intensity distribution. The mean images suggested that upper-body categories such as T-shirt/top, Pullover, Coat and Shirt have similar low-resolution silhouettes. This produced an initial hypothesis that these classes would be more difficult to distinguish than structurally distinctive categories such as Trouser, Bag or footwear.

Each 28x28 image was subsequently flattened into a 784-dimensional feature vector. Pixel intensities were normalized from the original [0,255] range to [0,1]. Scaling provides a consistent numerical representation for PCA and models whose behaviour depends on feature magnitudes or distances.

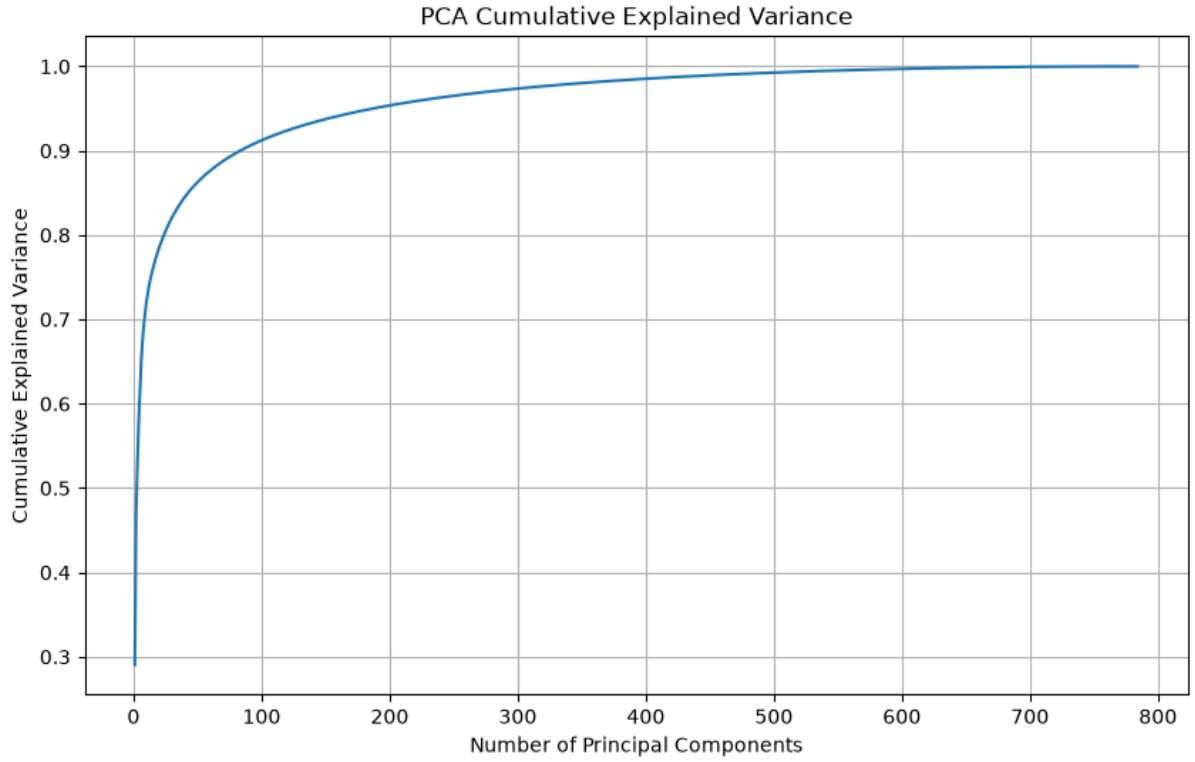
3. Experimental Protocol and PCA

The official 10,000-image test set was preserved for evaluation. The official 60,000-image training set was divided using a stratified 80/20 split, producing 48,000 model-fitting images and 12,000 validation images. A random seed of 42 was used for reproducibility.

PCA was fitted exclusively on the 48,000-image fitting subset to prevent information from validation or test data influencing the learned feature representation. The cumulative explained-variance curve was then examined.

The analysis showed that 84 components retained 90% of variance, 187 retained at least 95%, and 459 were required for 99%. The project selected the minimum number required to retain at least 95% cumulative explained variance: **187 components**. This reduced dimensionality from 784 to 187 features, a reduction of approximately 76%, while retaining most variation in the original images.

The same fitted PCA transformation was applied to validation and test data. All primary classifiers therefore operated on an identical 187-dimensional representation.



4. Baseline and Model Selection

Logistic Regression provided the Checkpoint-1 baseline. Using the 187-component PCA representation, it achieved **84.51% accuracy** on the official Fashion-MNIST test set. Its measured fitting time was 42.8113 seconds and test prediction time was 0.0064 seconds.

The four principal classifiers were subsequently tuned using validation data.

For KNN, k values of 3, 5, 7, 9 and 11 were compared. k=5 produced the highest validation accuracy at **86.62%** with macro-F1 **86.58%**.

For the RBF SVM, C values 1, 10 and 100 were combined with gamma values scale, 0.01 and 0.001. Because exhaustive RBF-SVM fitting on the complete training set is computationally expensive,

hyperparameter search used a stratified 15,000-example fitting subset with 1,500 examples from each class. The best search configuration was C=10, gamma=scale, producing **89.13% validation accuracy**. This configuration was subsequently trained using the complete 48,000-example fitting subset.

The selected Random Forest contained 200 estimators with maximum depth 40. It achieved approximately **86.96% validation accuracy** during tuning.

The selected MLP used two hidden layers containing 128 and 64 units with tanh activation. Initial experiments set max_iter=100, but the optimizer reached this ceiling before satisfying its convergence criterion. Rather than ignoring the warning, the maximum was increased to 200. In the convergence-check run, the network converged naturally after **96 iterations**, achieving **89.375% validation accuracy** and **89.403% macro-F1**. Therefore, 200 was retained as an upper bound rather than a requirement that every training run execute exactly 200 iterations.

5. Comparative Results

After hyperparameter selection, the four classifiers were evaluated using the common PCA representation and official test set.

Model	Test Accuracy	Macro-F1	Fit Time (s)	Predict Time (s)
RBF SVM	**89.91%**	**89.90%**	58.07	20.22
MLP	87.58%	87.59%	26.38	**0.013**
KNN	85.79%	85.74%	**0.030**	1.07
Random Forest	85.75%	85.55%	15.53	0.092

The RBF SVM achieved the strongest predictive performance, improving official-test accuracy by approximately 5.4 percentage points over the Logistic Regression baseline. However, it also had by far the greatest prediction cost, requiring approximately 20.22 seconds for the test set.

KNN demonstrated the opposite computational behaviour. Its fitting stage was almost instantaneous because KNN primarily stores the training representation rather than learning a conventional parametric decision function, but prediction was considerably slower than MLP or Random Forest.

The MLP provided the strongest alternative trade-off. Although its 87.58% test accuracy was below the SVM's 89.91%, its prediction time was only approximately 0.013 seconds in the recorded run.

These results demonstrate why classifier selection cannot be based solely on accuracy. The appropriate model depends on whether predictive performance, fitting cost or inference speed is most important.

6. Class-Level Error Analysis

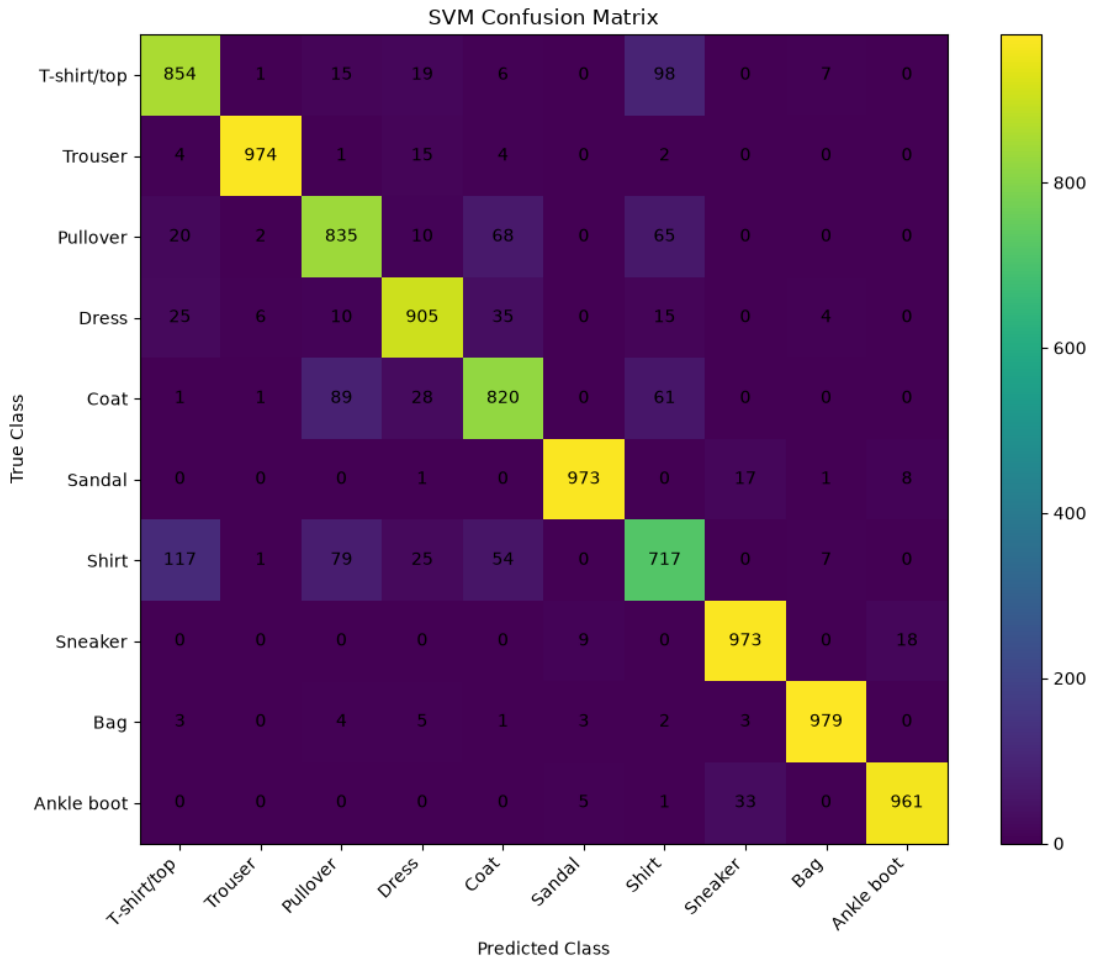
The SVM achieved an overall accuracy of **89.91%** and macro-F1 of approximately **89.90%**, but performance varied considerably between classes.

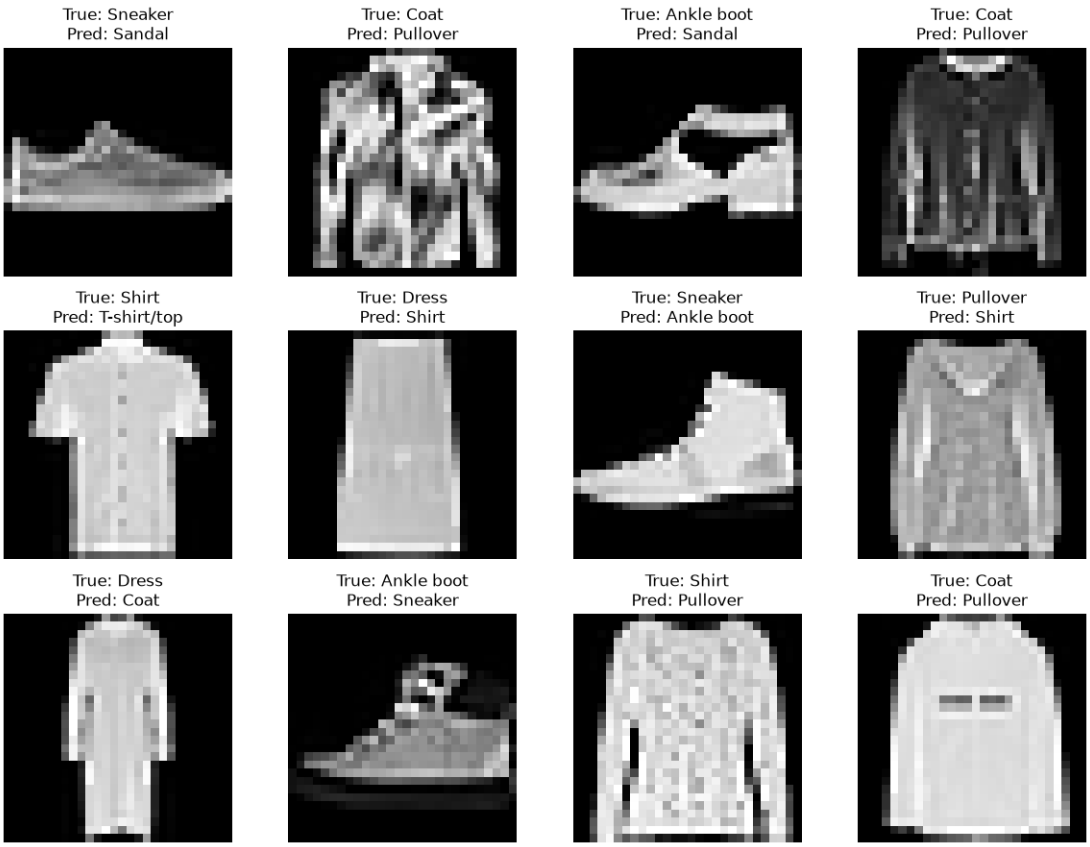
Shirt was clearly the most difficult category. Its precision was **74.61%**, recall **71.70%**, and F1-score **73.13%**. The largest confusion occurred between Shirt and T-shirt/top: **117 true Shirt images were predicted as T-shirt/top**, while **98 T-shirt/top images were predicted as Shirt**.

Other substantial errors were concentrated within the same upper-body group. Eighty-nine Coat images were classified as Pullover, 79 Shirt images as Pullover, 68 Pullover images as Coat, 65 Pullover images as Shirt, 61 Coat images as Shirt and 54 Shirt images as Coat.

In contrast, distinctive classes performed substantially better. Trouser achieved approximately 98.14% F1, Bag 98.00%, Sandal 97.79%, Ankle boot 96.73% and Sneaker 96.05%. A smaller footwear ambiguity remained, with 33 Ankle boots classified as Sneakers.

These findings support the hypothesis formed during exploratory analysis. Pixel-derived PCA features can represent overall shape and intensity structure but do not explicitly encode semantic properties such as texture, fabric, drape or garment construction. Consequently, visually similar upper-body garments occupy more difficult decision regions.





7. Effect of PCA Dimensionality

To examine the relationship between dimensionality and classification performance, the selected RBF SVM (C=10, gamma=scale) was evaluated using PCA representations containing 10, 25, 50, 100 and 200 components. PCA was fitted only on the model-fitting subset for each experiment, and accuracy was measured using the validation set.

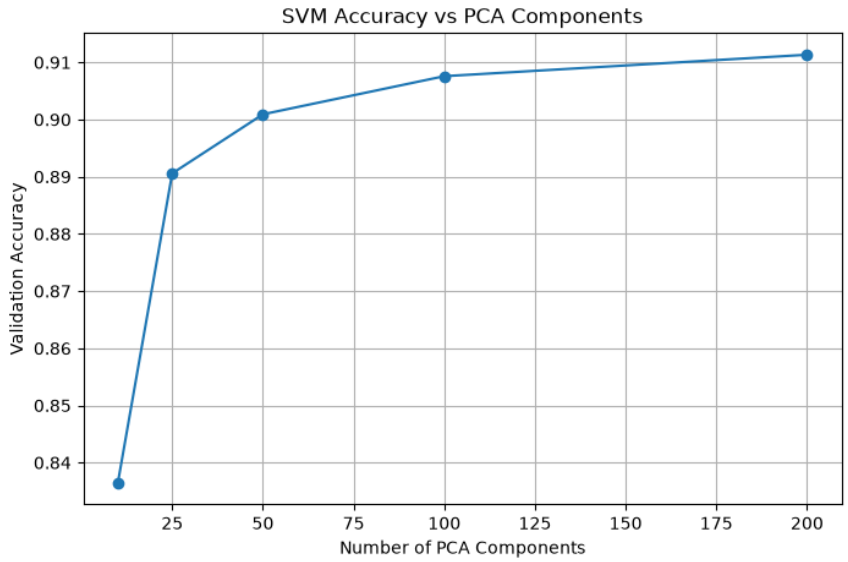
PCA Components	Validation Accuracy
10	0.836417
25	0.890583
50	0.900917
100	0.907583
200	0.911333

The largest improvement occurred between 10 and 25 principal components. Increasing the representation from 25 to 50 components produced a further improvement, but subsequent gains became progressively smaller. Accuracy continued to increase through 100 and 200 components, although the curve increasingly flattened.

These results demonstrate diminishing returns from additional PCA dimensions. The first principal components capture the most important variation in the Fashion-MNIST images, while later

components provide increasingly small improvements in classification performance. Beyond approximately 50–100 components, substantially increasing dimensionality yields only modest additional validation accuracy.

The project's primary 187-component representation was selected independently using the predefined 95% cumulative explained-variance criterion, rather than by optimizing this sweep against validation accuracy. The sweep therefore provides a separate analysis of the relationship between dimensionality and downstream classifier performance.



8. Limitations and Conclusion

The strongest model, RBF SVM, achieved **89.91% official-test accuracy**, demonstrating that classical machine-learning techniques combined with dimensionality reduction can classify Fashion-MNIST effectively without convolutional or other deep image architectures.

However, several limitations remain. Fashion-MNIST consists of standardized 28x28 grayscale images and therefore represents a much simpler visual environment than ordinary photographs. Performance on Fashion-MNIST should not be interpreted as equivalent performance on photographs containing backgrounds, varying lighting, orientation, scale or multiple objects.

PCA also compresses pixel-level variation rather than learning semantic visual features. This contributes to persistent confusion among Shirt, T-shirt/top, Pullover and Coat.

The project nevertheless demonstrates a complete classical image-classification workflow: binary data loading, exploratory analysis, scaling, dimensionality reduction, validation-based hyperparameter selection, comparison of multiple classifier families, computational timing and class-specific error analysis.

Future extensions can investigate targeted feature weighting, simple data augmentation, a dedicated Shirt-versus-T-shirt classifier, or a NumPy implementation of the MLP. An interactive image-upload interface could additionally demonstrate inference on user-supplied images, although such photographs represent a distribution shift from Fashion-MNIST and should not be included in the official test-performance claim.